

Computer vision

Noise

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Uncorrelated noise is defined as the **random intensity variations** within an image that have no spatial dependencies from image to image.

Spatial and Frequency Properties of Noise

Spatial Properties of Noise

Characteristic of noise in spatial domain is based upon:

- the **shape of probability density function** (analog case of noise)
- the **shape of histogram function** (discrete case of noise)

Frequency Properties of Noise

- Noise often spans a broad range of frequencies.
- For white noise, the power spectral density (PSD) is equal in each frequency.
- Periodic Noise

Noise – additive component

Model of degraded image by additive noise :

$$d = l + n$$

Where:

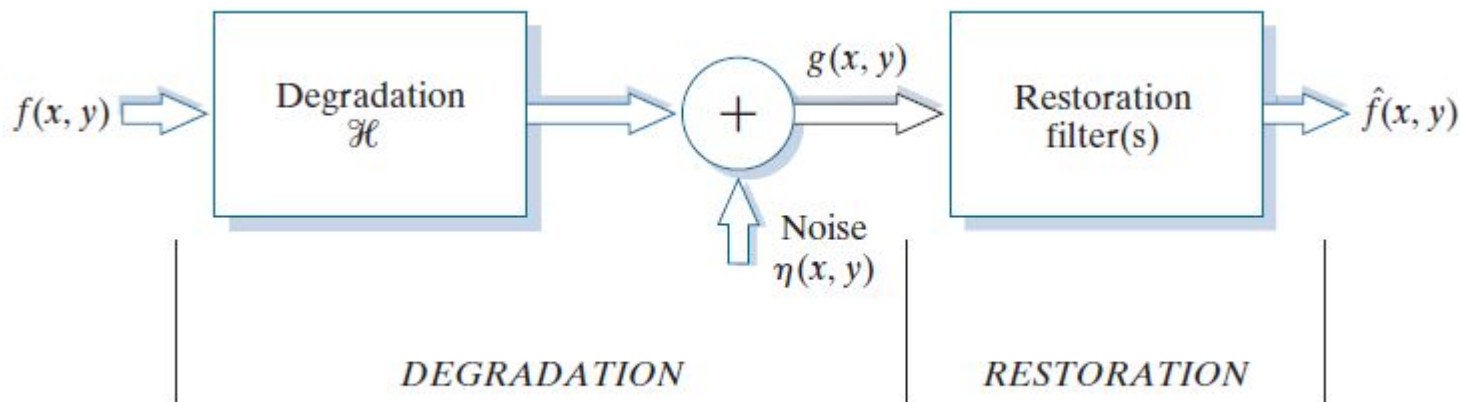
d is degraded image

l is original image

n is additive noise function

Noise Models

A model of the image degradation/ restoration process.



Typical image noise models

According to the shape of the histogram function, typical image noise models are:

- Uniform
- Gaussian (normal)
- Salt-and-Pepper (impulse)
- Gamma noise
- Rayleigh distribution
- ...and more

Uniform noise

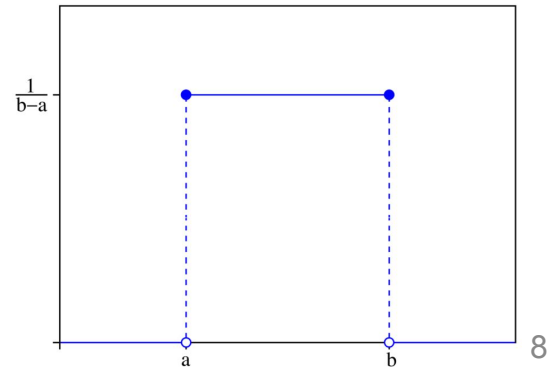
Uniform noise can be analytically described by:

$$\text{HISTOGRAM}_{\text{Uniform}} = \begin{cases} \frac{1}{b-a} & \text{for } a \leq g \leq b \\ 0 & \text{elsewhere} \end{cases}$$

$$\text{mean} = \frac{a+b}{2}$$

$$\text{variance} = \frac{(b-a)^2}{12}$$

Histogram of Uniform noise:



Example of Uniform Noise

Original image
noise and histogram:

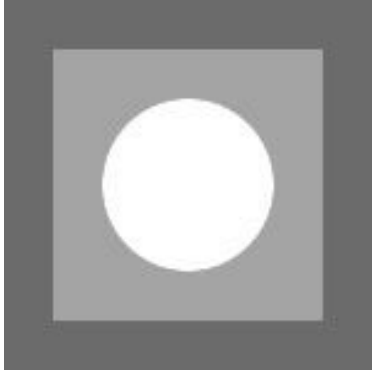
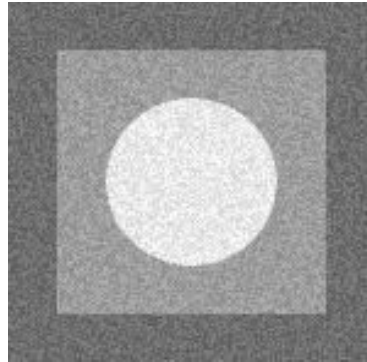
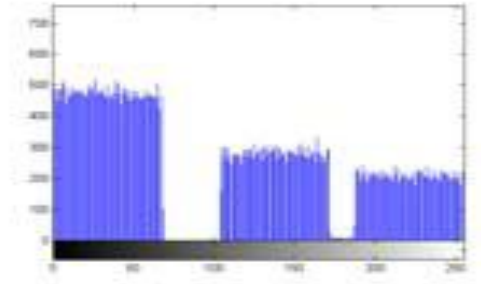
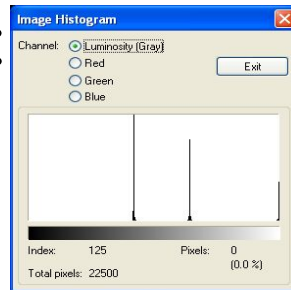


Image disturbed by uniform
and histogram:



and histogram:



Uniform noise

Quantization noise has an approximately uniform distribution.

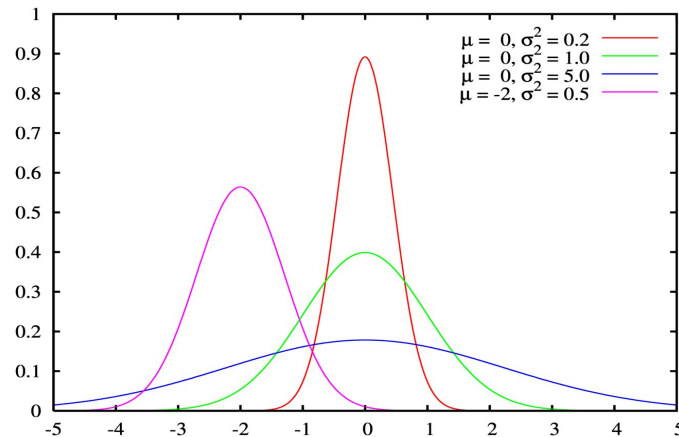
Uniform noise can be used to generate any other type of noise distribution, and is often used to degrade images for the evaluation of image restoration algorithms since it provides the most unbiased or neutral noise model.

Gaussian noise (Amplifier noise)

... is statistical noise that has a probability density function (pdf) of the normal distribution (also known as Gaussian distribution).

... is a major part of the "read noise" of an image sensor, that is, of the constant **noise level in dark areas** of the image.

PDF (Probability density function)



Example of Gaussian Noise

Original image
and histogram:

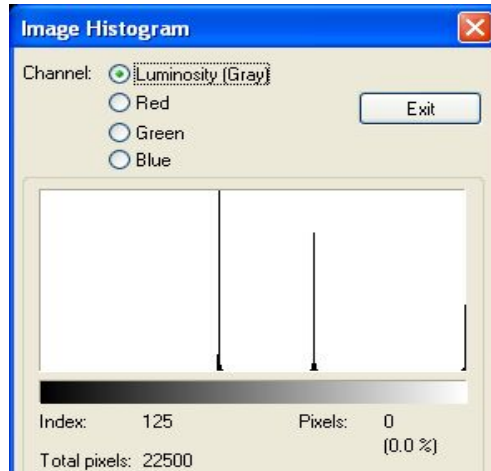
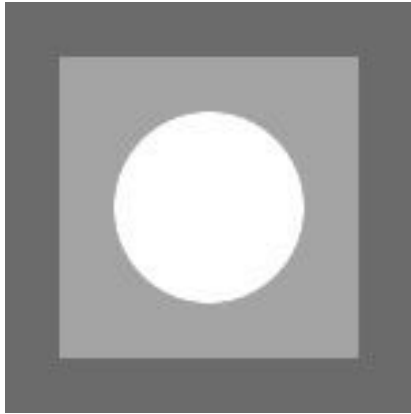
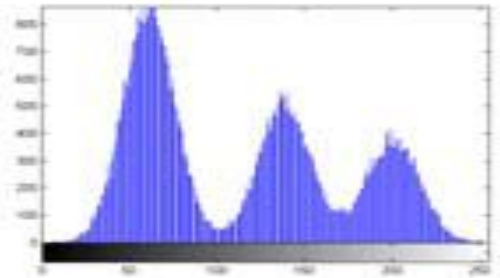
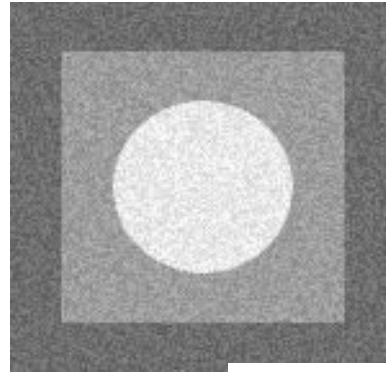


Image disturbed by Gaussian
and histogram:



Example of Gaussian Noise

Original image

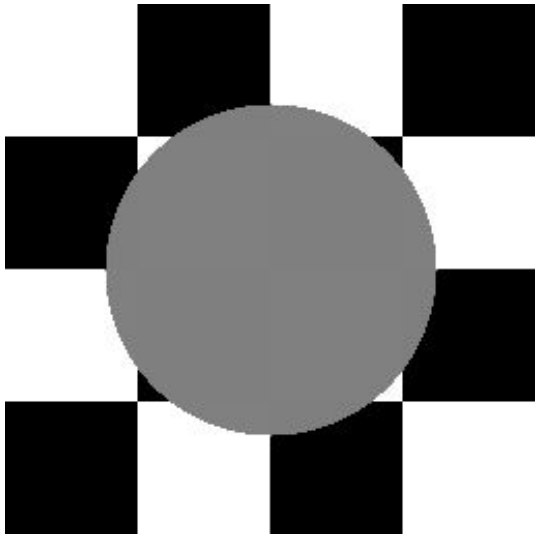
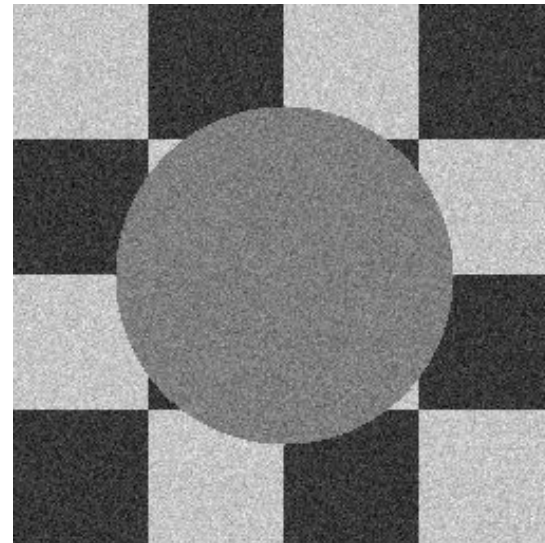


Image with added Gaussian noise



Salt and Pepper noise

Salt and Pepper noise can be analytically described by:

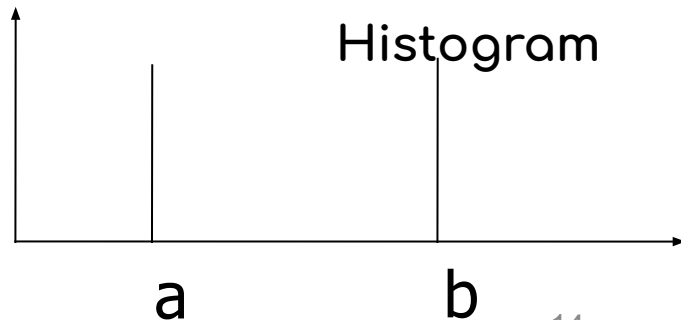
Histogram of Salt and Pepper noise =
$$\begin{cases} A & \text{for pixel val.} == a \quad (\text{"pepper"}) \\ B & \text{for pixel val.} == b \quad (\text{"salt"}) \\ 0 & \text{otherwise} \end{cases}$$

There are only two possible values, a and b.

For an 8-bit image,

the typical value for pepper-noise

is 0, and 255 for salt-noise



Salt and Pepper noise

The salt-and-pepper type noise (also called impulse noise, shot noise or spike noise) is typically caused by

- bad functioning pixel elements in the camera sensors,
- faulty memory locations,
- or timing errors in the digitization process..

Examples of Uniform Noise and Salt-and-pepper Noise

Image with added uniform noise

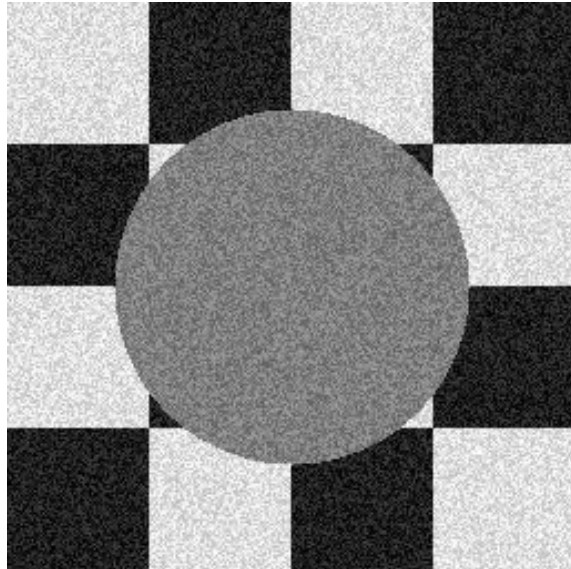
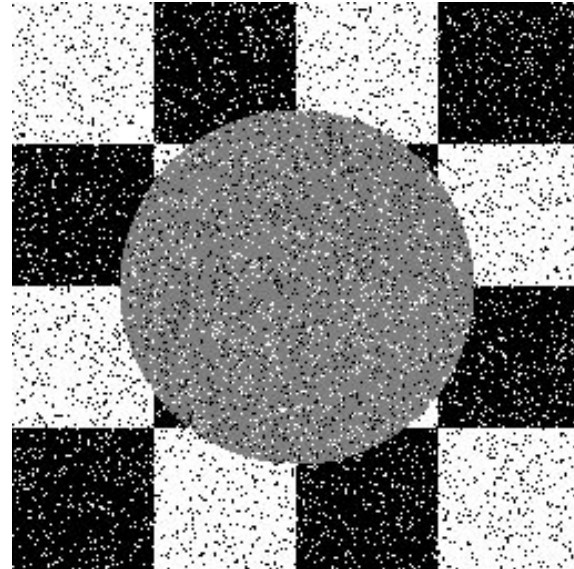


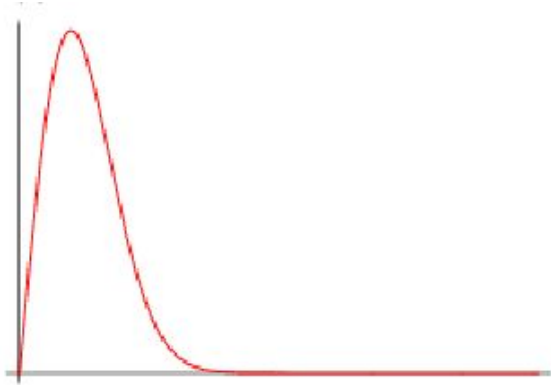
Image with added salt-and-pepper noise



Rayleigh noise

PDF (Probability density function)

Rayleigh distribution is defined as:



$$HISTOGRAM_{Rayleigh} = \frac{2g}{\alpha} e^{-g^2/\alpha}$$

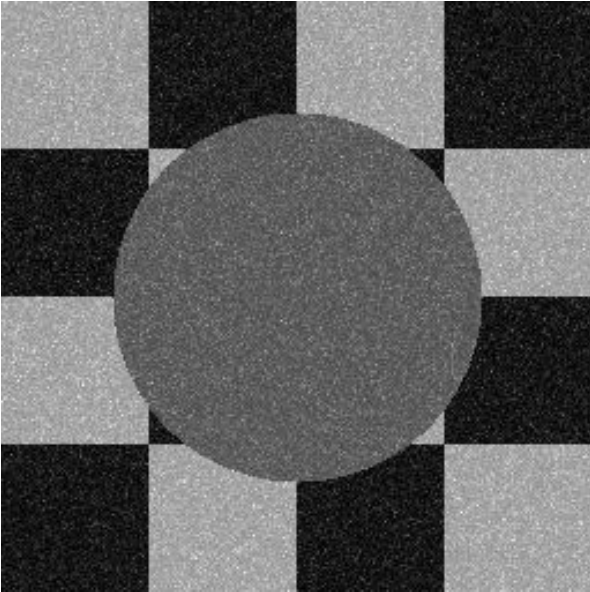
$$\text{where : mean} = \sqrt{\frac{\pi \alpha}{4}}$$

$$\text{variance} = \frac{\alpha(4 - \pi)}{4}$$

Radar range and velocity images typically contain noise that can be modeled by the Rayleigh distribution

Examples of Rayleigh noise

Image with added Rayleigh noise



Estimation of Noise

Consists of finding an image (or sub-image) that contains only noise, and then using its histogram for the noise model

Noise only images can be acquired by aiming the imaging device (e.g. camera) at a blank wall

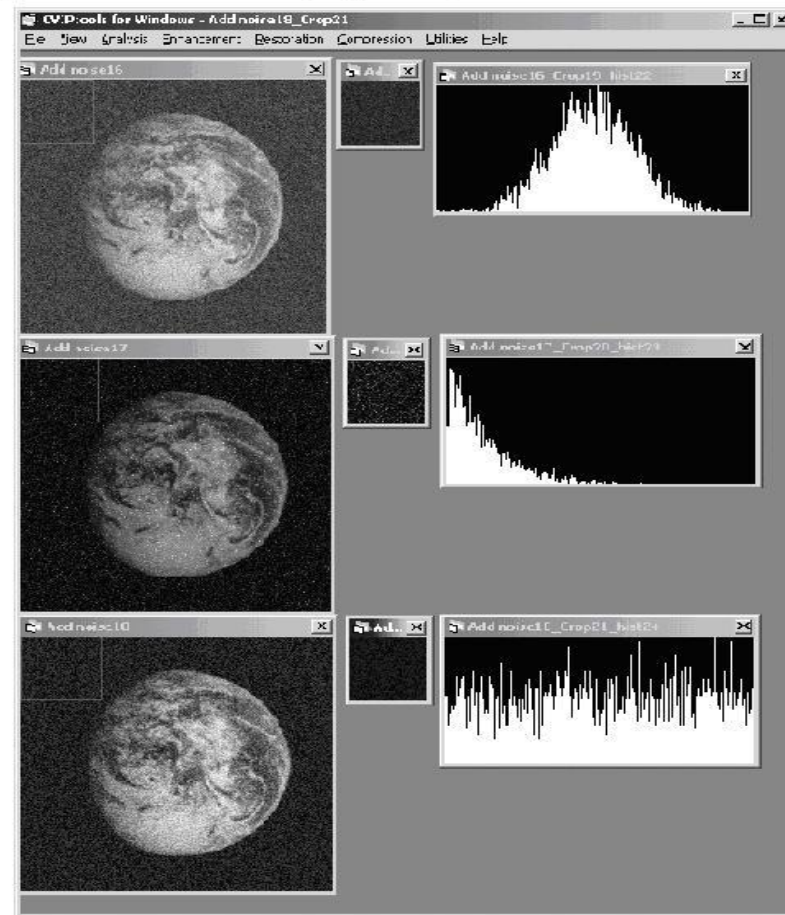
A "noise-only" portion of the image is selected that has a known histogram, and that knowledge is used to determine the noise characteristics

After a portion of the image is selected, we subtract the known values from the histogram, and what is left is our noise model

To develop a valid model many sub-images need to be evaluated

Estimating the Noise with Crop and Histogram

On the left are three images with different noise types added. The upper left corner is cropped from the image and is shown in the middle. The histogram for the cropped subimage is shown on the right. Although the noise images look similar, the histograms are quite distinctive – Gaussian, negative exponential and uniform



Methods of noise reduction

Spatial / frequency **filter**

Specific noise -> tailor-made filter...

The two primary categories of spatial filters for noise removal are

Order filters (Median filter...)

Linear filters (Low-Pass filter...)

Advanced methods :

Neural Networks

Noise in frequency domain

White noise

In signal processing, white noise is a **random signal** having equal intensity at different frequencies, giving it a constant power spectral density (PSD)

- White noise has zero mean, constant variance, and is uncorrelated in time.
- White noise has a power spectrum which is uniformly spread across all allowable frequencies.
- Infinite-bandwidth white noise signal is a theoretical abstraction

Periodic Noise

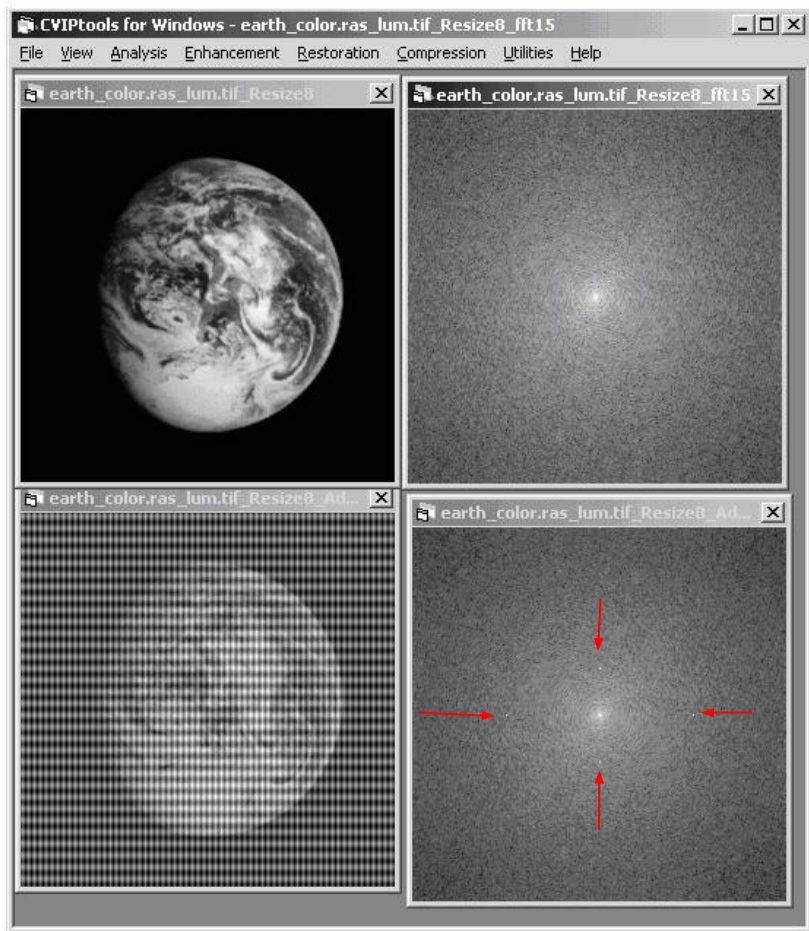
Periodic noise in images is typically caused by electrical and/or mechanical systems, such as mechanical jitter (vibration) or electrical interference in the system during image acquisition

It appears in the frequency domain as peaks corresponding to the periodic noise interference

It can be removed with band reject and notch filters

Image Corrupted by Periodic Noise.

On the top are the original image and its spectrum; under it are the image with additive sinusoidal noise, and its spectrum. Note the four impulses corresponding to the noise appearing as white dots – two on the vertical axis and two on the horizontal axis



Filtering of high frequency content

In most noiseless images the spatial frequency energy is concentrated in the low frequencies

In an image with added noise or noisy images, much of the high frequency content is due to noise

-> This information is useful in the development of models for noise removal

DEMO!